A Marriage Of Taylor And Fourier Series To Generate Definite Integrals

DR. TOM OSLER

JAMES ROSADO

MATHEMATICS DEPARTMENT
ROWAN UNIVERSITY
GLASSBORO, NEW JERSEY

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ABSTRACT:

In this presentation, we will demonstrate an unusual method of obtaining closed form solutions to complicated integrals of the form:

$$\int_{-\pi}^{\pi} f(x)dx \qquad \int_{-\pi}^{\pi} f(x)\cos(nx)dx \qquad \int_{-\pi}^{\pi} f(x)\sin(nx)dx$$

Where n is a positive integer.

This method is unusual because the integrals are the result of a process which does not envision the integrals initially; instead, the closed form integral solutions are the result of **comparing** coefficients of power series and related Fourier Series.

INTRODUCTION

We will utilize the…

- Fourier Series,

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx)$$

- Taylor Series,

$$g(z) = \sum_{n=0}^{\infty} c_n z^n$$

- and a complex function g(z)

To find closed form expressions to definite integrals. Some of these integrals maybe new.

We will demonstrate this methodology by an example.

AN EXAMPLE

Let $g(z) = e^z$ and substitute $z = re^{i\theta}$. We can express our function *g(z)* as the sum of a real and imaginary part:

$$e^{(re^{i\theta})} = e^{r\cos\theta + ir\sin\theta}$$

$$= e^{r\cos\theta} e^{ir\sin\theta}$$

$$e^z = \underbrace{e^{r\cos\theta}\cos\left(r\sin\theta\right)}_{u(r,\theta)} + \underbrace{ie^{r\cos\theta} r\sin\left(r\sin\theta\right)}_{v(r,\theta)}$$

REAL PART **IMAGINARY PART**

Let us make the same substitution for the Taylor Series expansion of our function g(z) = $\sum_{n=0}^{\infty} \frac{1}{n!} z^n$

$$e^{(re^{i\theta})} = \sum_{n=0}^{\infty} \frac{1}{n!} (re^{i\theta})^n = \sum_{n=0}^{\infty} \frac{r^n}{n!} e^{in\theta}$$

$$= 1 + \sum_{n=1}^{\infty} \frac{r^n}{n!} \cos n\theta + i \sum_{n=1}^{\infty} \frac{r^n}{n!} \sin n\theta$$

Observe that the line above is a Fourier Series expansion.

If we equate the results of the pervious two slides, we achieve the sine and cosine series expansions of the real and imaginary parts to g(z).

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$e^z = 1 + \sum_{n=1}^{\infty} \frac{r^n}{n!} \cos n\theta + i \sum_{n=1}^{\infty} \frac{r^n}{n!} \sin n\theta$$

$$e^z = e^{r\cos\theta} \cos(r\sin\theta) + i e^{r\cos\theta} \sin(r\sin\theta)$$

This is the Cosine Series Expansion of u(r,θ)

This is the Sine Series Expansion of v(r,θ)

Notice from the previous slide we have found Fourier Series expansions, but more importantly we have stumbled upon solutions to definite integrals. Recall:

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx \quad \Rightarrow \quad 1 = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\,d\theta$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx \quad \Rightarrow \quad \frac{r^n}{n!} = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\cos(n\theta)\,d\theta$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx \quad \Rightarrow \quad \frac{r^n}{n!} = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{r\cos\theta}\sin(r\sin\theta)\sin(n\theta)\,d\theta$$

We have achieved our goal of evaluating complicated definite integrals. With some simple manipulation of the three integrals, and keeping *'r'* fixed and θ is the variable of integration we obtain:

$$\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\,d\theta = 2\pi$$

$$\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\cos(n\theta)\,d\theta = \frac{\pi r^n}{n!}$$

$$\int_{-\pi}^{\pi} e^{r\cos\theta}\sin(r\sin\theta)\sin(n\theta)\,d\theta = \frac{\pi r^n}{n!}$$

This methodology works best if our original function g(z) can be conveniently decomposed to real and imaginary parts.

Using this method we have obtained solutions to 36 very complicated integrals:

$$\int_{-\pi}^{\pi} \cos(r\cos\theta)\cosh(r\sin\theta)d\theta = 2\pi$$

$$\int_{-\pi}^{\pi} \cos(r\cos\theta)\cosh(r\sin\theta)\cos(2n\theta)d\theta = \frac{(-1)^n \pi r^{2n}}{(2n)!}$$

$$\int_{-\pi}^{\pi} \sin(r\cos\theta)\sinh(r\sin\theta)\sin(2n\theta)d\theta = \frac{(-1)^{n+1} \pi r^{2n}}{(2n)!}$$

These integrals occur in groups of three, and some *maybe* new integrals